

ActInf Textbook Group ~ Cohort 3 ~ Meeting 11

(Chapter 5, part 2)

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hello it is uh April 5th and we're in our second discussion of chapter five in cohort three so this is going to be our final recorded session for the first half of the textbook chapters one through five and then next week we'll have an unrecorded session we will have time to to fill out the feedback form for those who haven't yet we'll discuss project ideas which people can also add ideas to we'll just talk a little bit more broadly and informally about the textbook and about the textbook group and then we will have a inter-semester break or it's not a semester but uh uh intermission and we'll pick up when cohort 4 begins we'll pick up with a second half of cohort 3 going through just as previous cohorts did chapter 6 7 8 9 10 and so on so we're in our second discussion of chapter five I'll head over there now and we'll begin just with anybody in the chat or via raising their hand or just unmuting how does chapter five sit with you what are any thoughts or questions about chapter five that you want to just raise at the outset Jonathan yeah I sort of feel that um while in some sense it says that you don't need to have digested chapter four to understand chapter five it still feels like you know there's a lot of background there which which is really important and the the lack of technical Clarity for from chapter four I sort of feel there's a bit of a hangover um from that into chapter five the technical debt hangover yes okay thank you Maria I was a bit frustrated with this chapter because I thought it would be more insightful but it seemed to me it was just more the same bunch of equations narrated and plotted I guess the um the apologists view would be chapter four was equations and plots in general and chapter five in contrast is applying it to the specific mammalian nervous system examples and to empirical studies however broadly I totally agree with that any any other just thoughts or primary Reflections on chapter five any aspect of it or some part of the discussion or even any other active inference learnings over the last week or two come to your attention when you're in this setting foreign right in the chat Terry you wrote I was struck by the similarity to neural networks you could describe a little more if you want to unmute or we can see how that might come into play in the previous chapter 10 discussion

from cohort three we were playing with the idea of reading chapter 10 first to provide a lot of the conceptual scope and the historical connections and similarities and dissimilarities of active inference with other topics and so like it's interesting to note neural network is described in the citations and twice in chapter 10 in this paragraph but there's a lot of context that is kind of held out till chapter 10. but the intuitions that you're having along the way are um very good any other just general thought or a question on five otherwise we will look to the questions and um address those which we haven't addressed previously continue on through so continue on this neural network just while we're here let's use the figure 5.5 so here we have 5.5 up the cortical system the limbic or the midbrain system associated with policy selection the spinal motor execution Arc using this differential between perceptions and set points and so then Terry you wrote the series of nodes passing information along a chain with inhibitory mechanisms alongside yet neural networks use Transformers while active inference uses free energy well great topic to go deeper into the relationship between neural networks and free energy principle and active inference models in live stream 51 I'm just going to the slides now the core claim is that here on the right we see the Markov blanket and the internal and the external States sense coming in perception action selection resulting in embodied action and uh an effect upon the generative process the niche which then provides observations so this is how we've been looking at it in the textbook on the left side we see in a three-layer an input layer one or more compute or intermediate layers which can have a wide range of connectivities and then some output layer so this could be the image comes in and the classifier label comes out or the classifier label comes in and the image generated comes out and then it acts in an environment which then provides another observation and in this paper they demonstrate that the variational free energy on this partition is equivalent to the loss function on this neural network so the training of this neural network is monotonically related to the variational free energy loss on this model learning the loss function is the function that's being minimized to update this neural network and so it's really quite a stunning conclusion because the topologies so how the pieces are connected of these two models is entirely different so in live stream 51 there's more detail on that um neural networks use Transformers while active inference uses free energy and Jonathan as Transformers in the post-2017 technical sense so in the sort of attention is all you need pre-trained Transformer timeline that we're on that's a good question and that feels specifically like a reinforcement learning framework or is it more General so I I meant there in terms of the the diagram you were giving where there was some neural network and a sort of feedback on an environment um yeah this

this one here which yeah it looks like a sort of RL um framework but are you saying this is sort of more more General than that it is more General As I understood the paper describing that for the observation sequence one to T and the parameterization all the weights of the neural network and then analogously the variational free energy of those very same observations and the parameterization of the base graph these two are monotonically related within a certain classes quite broad classes yes it was I I mean it's I believe a very major recent finding because it suggests that we can have this like this on one hand it looks like a circuit board and it's it's a lot to look at however there is a semantic interpretability here we see prior hidden State B that's how things change Through Time hidden State changing Through Time hidden state so there's hidden States changing Through Time here's the a matrix emitting observations or you could recognize decision making preferences guiding decision making so it's like there is an interpretability to this base graph and so the idea that a neural network which can have an arbitrary topology with absolutely no semantic interpretability kind of like any given neuron in a real neural system I mean Jennifer Aniston neuron aside no neuron plays an explicit semantic role let alone a one-to-one semantic role so the idea that this kind of an aggregate would have a representation where the variables are semantic is quite a claim and then the notion that the training function for both of them are monotonically related is really important not to go too far afield from the textbook but just neural networks are bought in and that's an example of some very recent research that makes that connection go beyond intuition okay here's a question that I don't think that we addressed previously but if anyone else has any other question please raise your hand or write in the chat so table 5.1 gives a remarkable association between neurotransmitters and precisions should we be suspicious that this seems so clean or is this evidence that the framework is a good match for real brain functioning what what does anyone think or what was their prior suspicion before seeing the table and then what was their posterior suspicion after they saw the table Jonathan and then anyone else yeah so this this was my question I I think I was just surprised that there was sort of this apparent one-to-one correspondent or at least near one-to-one correspondence so I think I would have expected in a different neurotransmitters to have you know a more um I guess just they're not to be this this very clean relationship um that was my prior ly these molecules do um things that are not only listed here like serotonin the word means blood pressure serotonin was discovered in the blood as was dopamine and it has a variety of effects on immune cells and then later was understood to play a role in the central nervous system so I don't think that these evidence provided are suggesting that the exclusive function of these neurotransmitter type

molecules is um that they're exclusively and exhaustively described by these evidences however these do provide empirical patterns of everything from histology to um association with cognitive functions neuroimaging studies um gene expression analysis pharmacological manipulation like the whole molecular Neuroscience toolkit these are quite disparate phenomena and their disparate phenomena that on one hand have been linked mechanistically to a given molecule and Associated receptors in all of this but on the other hand these phenomena are conceptually linked to certain to the semantics of certain parameters in these statistical models so maps are not territories it's not saying that it's exhaustively described if we use this Association as a heuristic if not a hypothesis and we find that we have unique explanatory and predictive powers then by the standards of science it was a valid modeling approach these are explanations and predictions that would not be achievable without mapping a molecule to the semantics of a parameter like just searching acetylcholine on a literature search engine you'll find a whole variety of different studies different study types different outcomes different species and so on so that's the um scruffy the scruffy is what the search engine just gives you each paper is unique every experiment is unique and then the the the neater the neat would be like all neurotransmitters do one thing and so somewhere that is useful enough to help us make explanations and predictions but not so terminally differentiated that we can't link together novel strands of evidence into explanations somewhere bridging or respecting the neat and Scruffy articulation is active inference and the ways that we can use statistical models to connect to these broad empirical results on different chemicals should we be suspicious that it's so clean well yes they're presenting their arguments like a prosecutor they're not necessarily going into every caveat or bringing up studies that might um disconfirm that so for sure it's connecting dots and not all the dots are being shown and not all the connections are being shown but it moves the discussion forward about how aspects of maps which are to say cognitive models are related to aspects of territory which include the actual materiality of the neural system Michael and then anyone else hello Daniel um sorry I've been um out of pocket for a week or so but um I'm trying to catch up in the way I'm trying to make sense of what I'm uh now Landing into I wanted to kind of validate does this make sense or not make sense um if if one were to think and the way I've been trying to the mental models that have been trying to map this conversation into are about social conversations and when you could say one level of activity up from if you will the biological biochemistry and I was thinking if these were uh verbs you know sorry if each of these um neurotransmitters were nouns verbs adjectives and that sentences were being created you

know they could be created many different ways but but that there are units of of composition if you will of of message signaling that are are creating larger sentences in in in at that level of the system would that metaphor make sense or would it be missing um how might it be a mismatch for what has happened great question does anyone have a thought on that potentially referencing figure 5.5 or any other area but any thought on this question oh I'll give a first thought then please anyone can build on it so what is speech really it's this continuous flow of like phonemes and and so it is not subject object verb in how it actually is like the materiality of speech is vibration in the air and all of that in the embodiment and so that's like the territory of speech but let's just say that we're putting on our linguist hat so now we're going to think about certain parts of speech as like subject object verb or noun an adverb and so on even though the speaker may have not thought about it that way we're making maps of sentences as linguists even if the speaker didn't use that map when they generated the speech let's think about what sentence does this whole figure represent and uh beliefs about this and beliefs about that lead to some differential contrasts in light of incoming evidence and the result of that now we're going to leave that little clause and the result of that contrast in light of incoming evidence is a decision-making process about what to given Precision on the decision-making process and one of the outcomes of that decision-making process a descending prediction that contrasts with incoming sensory data and results in an action hopefully it's not too mixed of a metaphor but absolutely these different compositions just like infinite grammar you can keep on adding yes and also in the social space you can keep on adding sentences to each other and make longer paragraphs you can keep on making longer sentences or lists similarly you can keep on making larger generative models or different generative models and so different connections amongst these units are going to be described kind of like by the genre like that was a haiku that was a sentence with no noun that was a question that was a directive so loosely it's a great working model to think about this as a graphically arranged sentence that's expressing something where some of the nodes some of the tokens are associated with more belief like States some of the nodes are associated with capacities for action and so on and so we can kind of like play with Legos play with language and compose like well what if there were these three sensory modalities and two action possibilities and decision making was done this way that's a generative model cool yeah and and then and then however that sentence is diagrammed it simply wouldn't be the actual speech itself it wouldn't be the vibration of the the voice box or the air so that's like map territory but once we're in the map then sentences might have this kind of structure or this

could be the dependency of an organization or the dependency structure of claims in a scientific Corpus that's kind of what Bayes graphs do they describe different variables and their associations and that also speaks to their flexibility like you could know all of these and solve for that one or you could just have these ones and try to reduce your uncertainty about those that'd be like a sentence where you did hear the first part but not the second part or vice versa thank you Dan again every time I join these conversations my mind gets mental models get blown open and thank you again good well hopefully they're not reduced to Rubble because your intuition was um right about taking a linguistic approach I mean who's to say it's right it certainly is compatible and um these this is like a composition um this is a setting where we can compose generative models and so that's why sometimes it's like a little challenge like what's the active inference um answer on attention what's the answer on memory it's like those are complex cognitive phenomena that play out differently in different systems so if the framework were rigid enough to give an answer to that in general it wouldn't have the latitude that we need to actually describe that phenomena and so we want to have the latitude to describe different phenomena including unnamed phenomena in systems we haven't seen yet which means that the framework itself can't be so specific about how certain things play out it's mind-blowing it's absolutely mind-blowing and it's such a problem I know I said every time it's such a privilege to be in this conversation and witnessing you make sense of it through this complex Collective lens when I've been thinking in blocks and um and these blocks are being you know it's like no these aren't bricks they're flows and and it works cause and effect works differently and it's just as I say mind-blowing and uh and daunting but thank you thank you it's definitely a team efforts and we've been through it many times in many people's insights have been folded together does anyone else in chat or raising their hand want to have any reflection on five or a question or we can look over some sections and then in the in the last minutes we can like look ahead to the second part of the book or or whatever people prefer is there a nerve a neural system or some embodied behaviors that people want to investigate I'll add in this question okay thank you Jana are there better or sorry are there alternative visualizations that better show the interdependencies that's really interesting question in a base graph the edges reflect yeah and I'll add this in thank you in the base graph the edges reflect a certain kind of statistical connection amongst variables but this entire structure it's kind of like [Music] a web or like a tensegrity sculpture or like a guitar string um in that the really an edge like let's just say it from this S3 to Pi 3. but yes but then if policy selection ends up doing something in the real world then that is going to influence future

proprioceptive inputs even edges that aren't shown it's not that they're not associated with each other through the reality of the system but the edges are showing the kind of direct connections again not the anatomical direct connections then it would just be an anatomy textbook but it's showing the direct informational or causal connections so how can we visualize the there's so many ways you might explore but I really like that as something for us to think more about like one more traditional way in terms of scientific visualization would be we could plot μ_{S3} as a on the x-axis and μ_{x1} on the y-axis and then we could ask whether there was any correlation or Mutual information between those two variables so the fact that they're not connected it doesn't mean it's going to be just a blur in fact they might be very strongly associated with each other but through μ some intermediate nodes so that would be one way of looking at whether to Any Given nodes two or three or however many whether they have associations and or interdependencies μ and then I guess another like little layer is like the space of the counterfactual like some interdependencies might not be revealed unless something changes in the so that is also a layer where two things might have no correlation and then you remove you remove an edge and then all of a sudden two things that weren't in association μ now are μ okay I I don't think that's a complete answer because there's a lot of work with visualization to be done but that is just kind of some aspect of it but then just continuing with the honest question that then to Jonathan so so it is like a magnetic coil the electric field is not going along the coil it is a field a lot to think about don't have a specific thought if anybody wants to add something but yes I like what if these are wires and then there's like fields of influence that aren't just the wires but the connections are there and they're relevant but then there might be other dependencies that aren't simply mapped out on The Wire dependencies okay Jonathan I'm going to copy your question in has this been looked at in hypograph language rather than graphs in active block fronts we we looked briefly at hypergraphs we haven't seen any active hypographs it's very interesting a graph Through Time so for example what if μ this right now there's a static topology so even if there's Dynamic information flowing through this graph you could say that it's a static graph the referent is dynamic it's changing through time but the structure is not changing Through Time but what if μ on even and odd time steps μ this connection existed or not then that kind of a of a system as far as I understand is μ a hypergraph is there some other aspect to hypergraph Jonathan or what would enabling this inactive inference help us do so hypographs sort of allow for more General types of edges μ it it sort of allows for edges which specifically go through vertices kind of the the easiest μ example is a is a plain routing

system airplanes where if a single airplane goes from um airport A to B to C um it's very clear that it's the same airplane that's gone between those and just by drawing an edge between A and B and B and C that doesn't give the same information that sort of a single piece of information or a singular single airplane has taken a particular route so it just allows for for more generalizations of um of edges than in in a regular graph I just wonder essentially about the the way you can encode the types of information that move through a um through a regular graph and a hypograph and one might be able to encode more specific information in a in a hypograph yeah very interesting I'm pretty sure that it hasn't been written on in active a lot of the papers seem very recent like the one you point to is um 2018 and the others are recent so it's um it'd be an interesting area um one note on this to kind of um speak to the other side where the kind of map uh is very effective with respect to the territory for better and For Worse what if these were computers and these were the literal wires connecting them in that case you could design in silico or cyber physical systems where the edges of the base graph very well mapped to the causal structure of the system physically the the kind of one of the hardest systems and one of the richest systems is like trying to make simple statistical models of the brain but when making just in silico systems so just simple pedagogical models or intuition pumps or in certain systems for for um digital and Communications systems the edges can have a much more clear interpretation because as laid out in the beginning of five this is kind of this is the tension and the dream would be what what if the actual anatomical connections amongst these cell types what if those were actually reflective of the structure of a base graph and and you'll note that they're not exactly exactly the same like just here's you know two to three and here we see that's two to three but also it goes from here from from um four to two but we don't see a four to two in the tissue so but but it's close enough but clearly not close enough to Merit calling this simply layer 5. and then I guess to the electric field these connections are the neural what if there are glial connections what if there are electric fields um influences like what the EEG measures are some of the electrical phenomena of these cells and actually just a specific kind of electrical phenomena of these cells the EEG outside of the head but those electrical signals are not only um outputs like Epi phenomenal to neural activity they're also inducing different changes in neural activity as well so that there's also physical edges that aren't shown just you know complex systems are complex and they engage in in multiple mechanistic ways which is why it's important to have maps of them Ali uh yeah uh well actually going back to Yana's question and also to uh touch on what Jonathan just uh mentioned uh well uh in fact when we're talking about uh interaction uh obviously there are

many different conceptions of this term interaction so we can talk about physical interaction which is basically uh I mean for the most part um is undertaken uh the fields so we can also term it as a kind of field interaction but there are some other conceptions of interaction actually there are many others but probably one of uh the most important conceptions for our purposes is information theoretic interaction uh so uh when we're talking uh talking about message passing or uh this conception of interaction through and I mean information passing or information theoretic conception of it we're not necessarily deciding on what's it what exactly is the medium of interaction or transmission because it can be uh abstracted away this medium of transmission or interaction and we can just look at the um I mean mathematical framework of um or the mathematical structure of those information so for instance um say uh IIT framework integrated information framework in which uh well in information integrated information Theory it's unable to distinguish a highly interdependent set of components dominated by a redundant relations from an equally strongly interacting set of components exhibiting a kind of strong Synergy between them so this is actually kind of a limitation for IIT for some people because since redundancy indicates that although a strong interaction is needed to ensure that a component is a kind of repetition of the other components it does not imply that the set of components support Collective emergent Behavior which is entirely different from that of the individual components so in recent years by recent I mean since 2020 there are some interesting Works in that regard trying to formulate uh the the interdependencies and emergence with mathematical formalism so Lionel Barnett if I remember his name correctly and also Anil Seth and others for custom of their names has done some pretty interesting work in that regard in recent years and one of the uh interesting things they've come across is that they've tried to apply this uh abstract conception of interaction and interdependencies from information theoretical theoretic point of view to neuroimaging data and uh so they and they've been able to describe this interaction of neural interaction between different brain regions without explicitly accounting for synaptic or I don't know a junction uh interaction or so on so it's much more abstract and much more generalized form of uh interaction awesome anatomy and physiology are different Anatomy is describing the articulations of the system but then the physiological relationships and correlations and even causes are not necessarily following along the anatomical lines if one wants the anatomical answer you can just slice the tissue and take the picture so something different is happening with this modeling and it's important like you mentioned that this interaction concept amongst variables by the time we're labeling variables we're in the

map we're not labeling the tissue anymore so the only possible kind of interaction that these variables can have is like statistical or information theoretic but then there's this tantalizing dream slash fallacy slash actionable framework all of those that parameters do map onto aspects of the territory and that's the map territory dialectic some maps useful in some contexts doesn't reify the the architecture of the map and it doesn't prove or disprove any architecture of the system and the interesting thing is that's not even active inference specific these are active inference type thank you Ali these are active inference or predictive processing like models any statistical or cognitive modeling approach is going to confront this issue and then how do we move Beyond just simply saying well map territory means we we can't drive or we can't do route planning it's like well if dopamine is associated with Precision in this model where should the deep brain stimulator go or what kind of food would make sense to eat or not eat or and so on so we can actually reduce uncertainty about what actions to take using Maps and that's what these are and those Maps arguably works best when it's understood what's a map and what's the territory and understanding what the map is can be fully described and held to the standards of of reproducibility whereas holding a territory to those same epistemic Norms or criteria it it doesn't even make sense the territory can't be fully described but when they're overlaid even symbolically it can become ambiguous next week we're going to not record so if you're going to be there again that's awesome otherwise please fill out the feedback form in the future textbook group section and check out the project ideas but in these last 10 minutes today is there any sections or questions on that anybody wants to look at or would people like to see into chapter six um I've just added those additional comments I was making into your coda and I'm making a quick visualization of what I was thinking about I'm not sure if this relates directly to the model um but just I'll need a minute to get this done thank you all I'll leave the tab up we'll return to it any preference for a section of five or to to look ahead to chapter six in the recipe even though we'll return to that in like a month looking at yes go ahead Michael I was going to say no objection here I just wanted somebody to respond okay let's revisit the whole first half of the book which you all have now completed one or more passes through chapter one provided an overview chapters 2 and 3 describe the low road and the high road the low road how are we going to do it with Bayesian statistics uncontentious why at the very least we need repeated measurement if something can't be repeatedly measured by us it's not a thing to us and then a little bit more from the inside if something can't take adaptive action or more adaptive action to persist and repeatedly measure itself it won't be that kind of thing either so whether you view from the outside repeated measurement

or you view from the inside adaptive action we can model both of those in terms of this self-evidencing primary imperative how are we going to model that self-evidencing primary imperative Bayesian statistics chapter four with all of its shakes and bumps was describing the generative models of active inference and how variational Bayesian inference is used as a heuristic or an approximation to otherwise intractable Bayesian statistical models and we just looked at which characterized a few subsystems of the mammalian neuroanatomy and then linked them together and provided some references to other models that go into a lot more detail computationally and and molecularly here's what is going to be coming up in the second half of the book that is the application oriented part of the book now it may not go all the way to the guidebook Playbook handbook that we all expect and prefer but it is more pragmatically an application oriented than the first half of the book in chapter 6 there's a recipe for Designing active inference models with some here we've even gone through that recipe but broadly they're going to lay out some steps about systems of interest about specification of the generative model and the generative process and there's a lot of good discussions around even broader considerations like why are we making this model who are we making the model for is this model being used descriptively like historically or is it being used proactively that's the recipe so you want to open up a restaurant here's the active inference recipe chapter 7 and chapter 8 are like a pair chapter 7 is going to cover generative models that have discrete time characteristics chapter 8 is going to look at octave inference in continuous time discrete time active inference on the top of figure 4.3 continuous time active inference generative model on the bottom of figure 4.3 so chapters 7 and 8 are going to basically zoom in on figure 4.3 in sequence chapter 9 model based data analysis is going to describe some of the framings and challenges and opportunities associated with integrating empirical data with these kinds of generative models so far up till then it's kind of been like here's the generative model we constructed and now we're going to simulate Its Behavior so generative model specification simulation in silico generation of synthetic data but we're not always just interested in generating synthetic data sometimes we want to take in empirical data from measurements and that's what this chapter is about and then chapter 10 is a longer chapter there's no equations or figures and it is a Capstone that's describing a lot of the conceptual and historical context and as kind of referenced earlier um may make sense to even read first that is where we're going in the second half of the cohort so the first half as the authors lay out is epistemic we're getting like the roads into the city what the city is about or one restaurant in the city and then a few dishes that have been prepared in this city then we're going to go behind the scenes look at the

recipe for Designing those talk about two subtypes of Cuisine the discrete time and the continuous time talk about connecting these generative models to measurements so that people can do analyzes and write papers and all of that make things that do things in the world and then having done all of that chapter connects a lot of dots that's the par at all textbook and for being here you all are at least half done with it and um as also mentioned we we go at a similar pace we haven't set the dates yet for the second half of cohort three it'll likely pick up again in May when we have cohort 4 begin with chapter one and um if you have any feedback or ideas it would be extremely appreciated to fill out this feedback form just directly in Coda you can just submit it right there as well as to share the link for others to join and convey to them just say hey stick around you will learn it's not XYZ or it is XYZ but feel free to share that so that we can have really engaged and awesome cohorts such as this one and also any other comments you want to write there or any thoughts on how we can improve it and next week um we're gonna Focus all about that we're gonna not record that session we're going to talk about the project ideas table overall Reflections on the first half next steps in about the future of the Jana I'll look forward to your image and in the last minute or two anyone else please feel free to add other comments it's not exactly goodbye as we will meet again but this is the last recording for this half of the cohort I'll uh put the image in Coda um in a few minutes so I'll I'll let you know Discord uh when it's ready thank you and it just makes me think about like these base graphs are kind of like the statistic this is the necessary and sufficient information for a statistical we might tell stories or link associations that don't follow this exact topology so this is not like the only web we can draw but it's a really important one for certain kinds of statistical models all right so next week same time for feedback and next steps thank you everybody see you then